KARTHIKEYAN P

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PROFESSIONAL SUMMARY

Generative AI Developer and Senior Analyst in AI/ML Computational Science with nearly 3 years of experience building and deploying enterprise-grade Large Language Model (LLM) applications. Proven success in designing advanced **Agentic RAG pipelines**, **multi-agent systems**, and deploying solutions on cloud platforms (**GCP, Azure, AWS**) to deliver measurable ROI. Seeking to apply deep expertise in **LLM architecture** and **Python** development to solve complex business challenges.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL
- **AI/ML Frameworks & Libraries:** LangChain, LangGraph, Agent Engine, Agent Development Kit (ADK), TensorFlow, PyTorch, Scikit-learn, CrewAI, NLP (GPT, Gemini, BERT, LLaMA)
- **Tools & Platforms:** AWS Bedrock, Gemini Enterprise, Docker, Streamlit, FastAPI, Vertex AI, Google Cloud Platform (GCP), Azure ML, Azure DevOps, SAP HANA, SAP Analytics Cloud, Git, GitHub
- **Databases & Vector DBs:** PostgreSQL, pgvector, Valkey/Redis, Pinecone, ChromaDB, Neo4j (Graph)
- **Core Concepts:** Retrieval-Augmented Generation (Agentic RAG), LLMOps, Model Fine-Tuning (LoRA), Multi-Agent Systems, Function Calling, Prompt Engineering, Role-Based Access Control (RBAC)

PROFESSIONAL EXPERIENCE

Accenture | Senior Analyst - AI/ML Computational Science | Nov 2025 – Present

FPG HR Assistant (Multi-Agent Conversational HR/IT Solution) (Mar 2026 – Present)

- **Spearheaded** development and deployment of a production-grade multi-agent conversational assistant utilizing **Python 3.13**, Google ADK, GCP (Agent Engine, Cloud Run), and **Gemini Enterprise** to automate HR and IT ticket workflows.
- **Architected** a modular multi-agent system with a Gateway Orchestrator and specialist agents for conversational context hand-offs, security, and strict domain enforcement of company policies and SuccessFactors updates.
- **Enhanced Reliability & Quality:** Implemented deterministic escalation guardrails and a robust ServiceNow ITSM connector supporting automated document interleaving. Drove a 100% test coverage target with over 630 automated pytest assertions and integrated SAST (Bandit) and styling checks (Ruff) for secure, high-quality code.
- **Leadership & Technical Mentorship:** Accelerated team adoption of new technologies (Gemini Enterprise, Agent Engine, eval frameworks) by creating clear, simple internal documentation and providing hands-on guidance.

IOCL Analytics Agent (Agentic RAG Platform) (Nov 2025 – Mar 2026)

- **Architected** an enterprise **Agentic RAG platform** using **LangGraph** and **AWS Bedrock** for natural language querying of complex market share data.
- **Engineered** a Hub-and-Spoke **dual-agent system** (SQL & Chart Agents) with a **PostgreSQL/pgvector** hybrid storage layer.
- **Implemented Valkey/Redis** caching and a dynamic context manager, enabling secure, instant **Plotly** visualizations and SQL-backed data tables.

Maventic Innovative Solutions | GenAI Developer | May 2022 – Oct 2025

- **Deployed** enterprise **LangGraph** and **FastAPI** chatbots, automating over 500 daily **SAP HANA** queries and reducing manual data retrieval for the sales team by **10+ hours per week**.
- **Led** the architectural design of a core **RAG pipeline** utilizing **Pinecone** and a **Neo4j knowledge graph**, improving technical document retrieval relevance by **40%**.
- **Built** predictive maintenance models achieving **95% accuracy**, which resulted in a **15% reduction** in unplanned equipment downtime.
- **Managed** end-to-end **MLOps** using **Azure ML** and **DevOps**, establishing CI/CD pipelines that reduced release cycles by **50%**.

PERSONAL & ACADEMIC PROJECTS

- **AI Research Analyst Agent:** Architected an autonomous research agent using LangGraph and Pinecone for multi-step web research and structured reporting.
- **Personalized Coding Co-Pilot:** Developed a context-aware coding assistant using a RAG pipeline (ChromaDB, Tree-sitter) by embedding an entire codebase for long-term memory.
- **Dynamic Customer Support Agent:** Designed a stateful LangGraph agent with custom tools and a dual-memory Redis system to query live SQL databases.

CERTIFICATIONS

Machine Learning (Stanford), Data Science (IBM), Prompt Designing (Google), Streamlit with Gemini in VertexAI (GCP), RAG with VectorDB and KnowledgeGraph (Deeplearning.ai), Function Calling and Data Extraction using LLM (Deeplearning.ai), LangGraph, LLMOps (Deeplearning.ai).

EDUCATION

Woolf University (Alma Better) | Remote
Master of Science in Artificial Intelligence and Machine Learning | 2024 – Present
CGPA: 8.7

KCG College of Technology | Chennai, Tamil Nadu
Bachelor of Engineering in Computer Science and Engineering | 2019 – 2023
CGPA: 8.5